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College of Engineering, New Delhi

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BHARATI VIDYAPEETH



Hon'ble Founder Dr. Patangrao Kadam

APPLIED PHYSICS - I

UNIT – IV (LASER)

- SYLLABUS : Spontaneous and stimulated emissions, Einstein's coefficients, Laser and its principle, He-Ne laser

Introduction

LASER

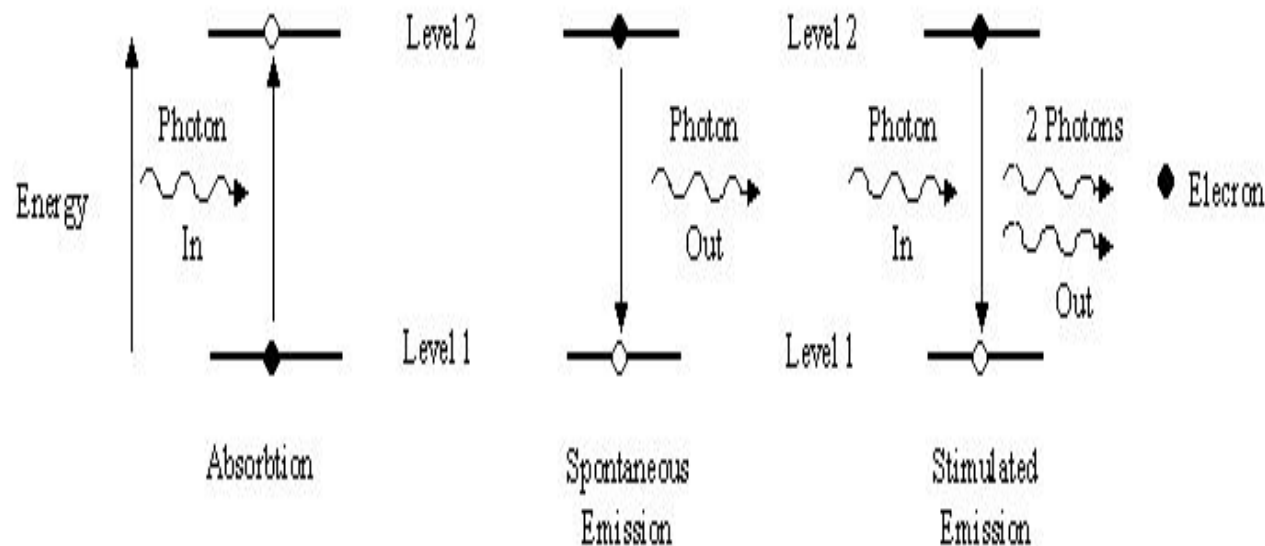
Light Amplification by Stimulated Emission of Radiation

Process of Laser

There are basically three phenomena

1. Stimulated Absorption
2. Spontaneous Emission
3. Stimulated Emission

Diagrammatic Representation



Einstein's Coefficients

- Consider two energy levels with energy E_1 and E_2 . If N_1 is the population of level 1 then

$$dN_1/dt = -B_{12} N_1 E(\nu)$$

Here B_{12} is the coefficient of stimulated absorption and $E(\nu)$ is the energy density.

Further $(dN_2/dt)_{\text{spn.}} = -A_{21} N_2$

A_{21} is coefficient of spontaneous emission

Further $(dN_2/dt)_{\text{st}} = -B_{21} N_2$

B_{21} is coefficient of stimulated emission.

- Relationship between Einstein coefficients is given by

$$B_{12} / B_{21} = 1 \quad \text{and} \quad A_{12} / B_{21} = [8 \pi h(\nu^3)] / c^3$$

The ratio of spontaneous and stimulated emission is proportional to ν^3 .

The probability of stimulated emission is the same as that of absorption.



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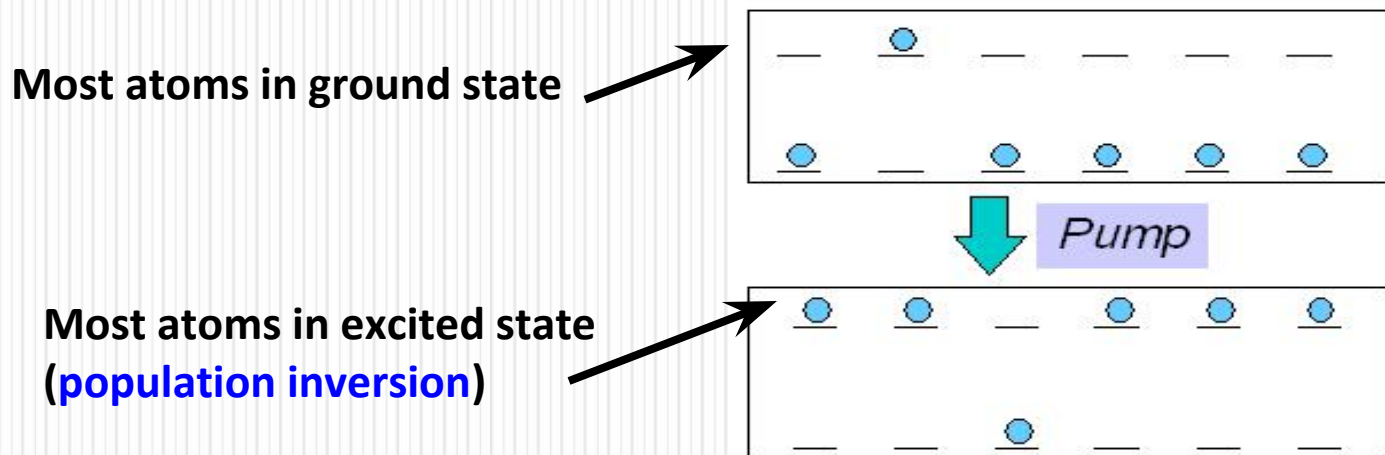
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Necessary conditions for a lasing transition

- Population inversion
- Metastable state

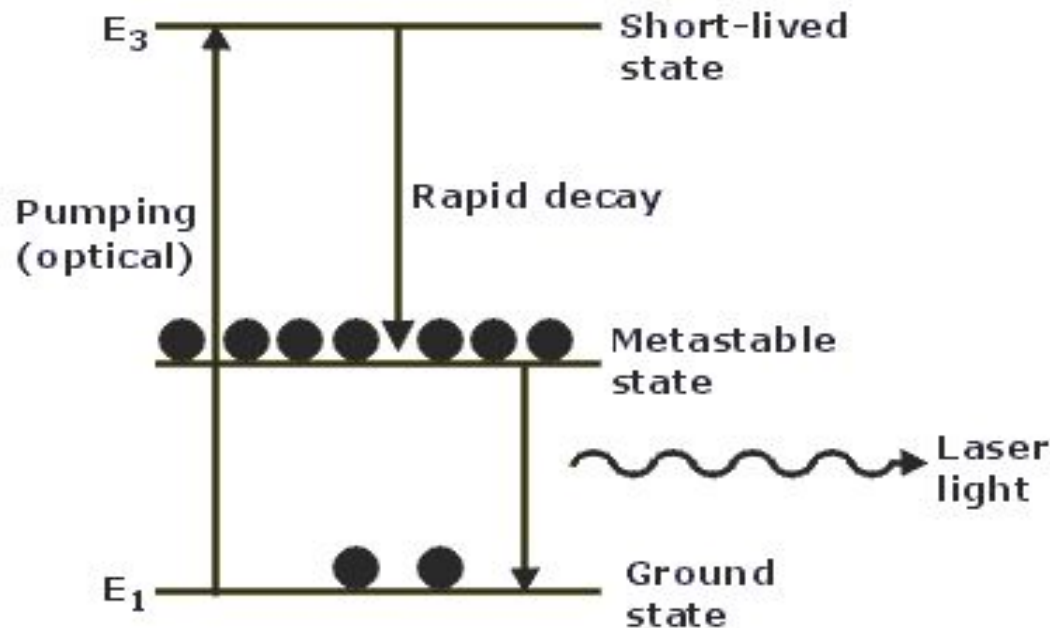
Population inversion

- **Stimulated emission** is the basis of the laser, but we need many more atoms in the excited state than the ground state. This is called **population inversion**. It is achieved by **pumping**.



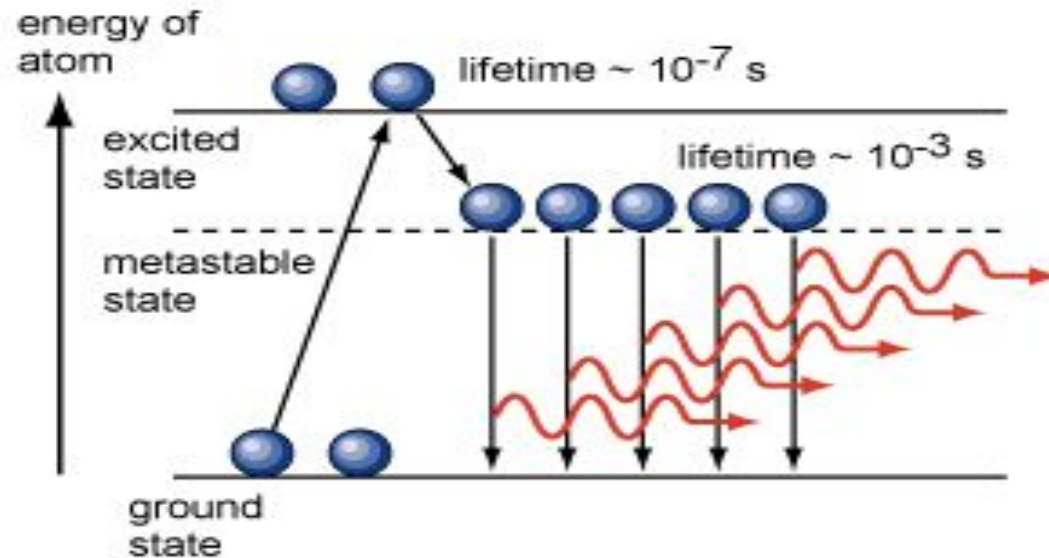
Metastable states

To achieve population inversion we must have **metastable states**. These are excited states where electrons stay for unusually long times.



Metastable State

Normally an electron in an excited state will make the transition to a lower state in a time of 10^{-7} s. In contrast an electron may stay in a **metastable state** for 10^{-3} s

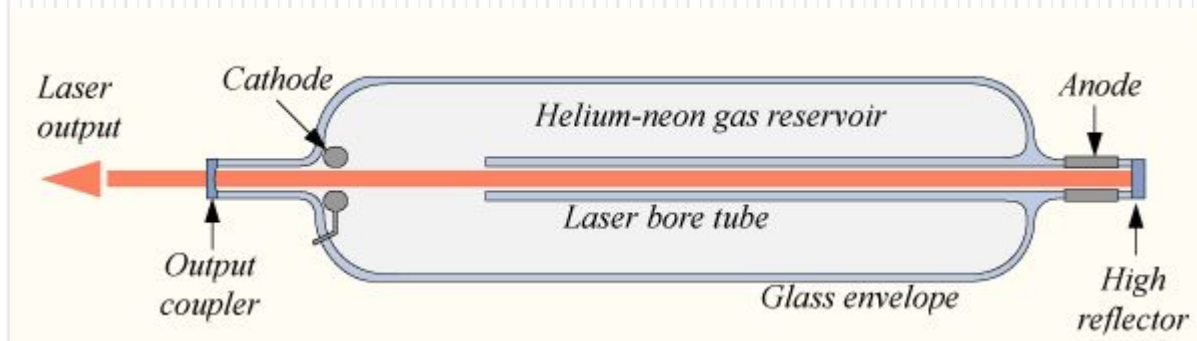


He-Ne Laser

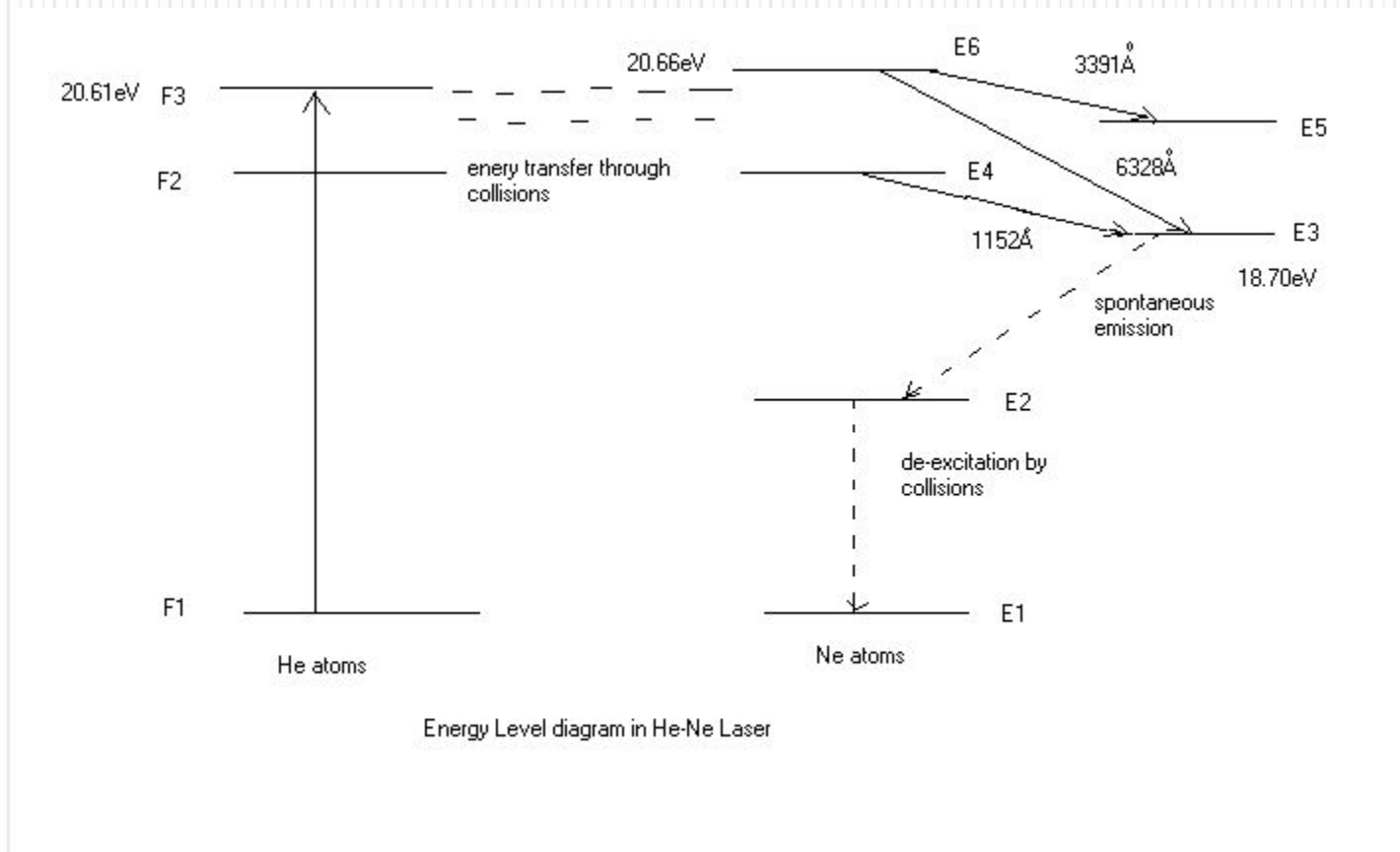
- The He-Ne laser was the first continuous-light gas laser, invented by Ali Javan in 1961.
- The He-Ne Laser is a four-level laser.
- Optical gain medium is He and Ne gas mixed in the ratio 10:1 and enclosed in a glass tube at pressure of 1Torr.

- wavelength of laser is 6328 Angstrom, which lies in visible region and it is of red colour.
- Its main use is in the telecommunication industry specifically through fibre optics technology

Construction.



Energy Level Diagram



Applications

- Fibre optics communication
- Medical Applications
- Laser Nuclear Fusion
- Barcode Scanners
- Heat Treatment
- Spectroscopy
- Laser Printing
- CD s and optical discs